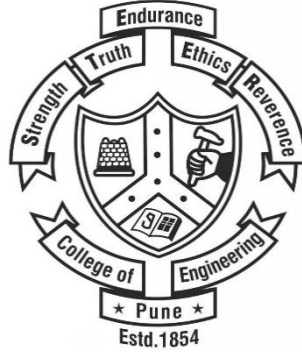




College of Engineering Pune

(An Autonomous Institute of the Govt. of Maharashtra)

YEAR : SECOND YEAR BACHELOR OF PLANNING
SEMISTER: IV

SUBJECT: Geo-informatics-I

Lecture 4

Prepared by,

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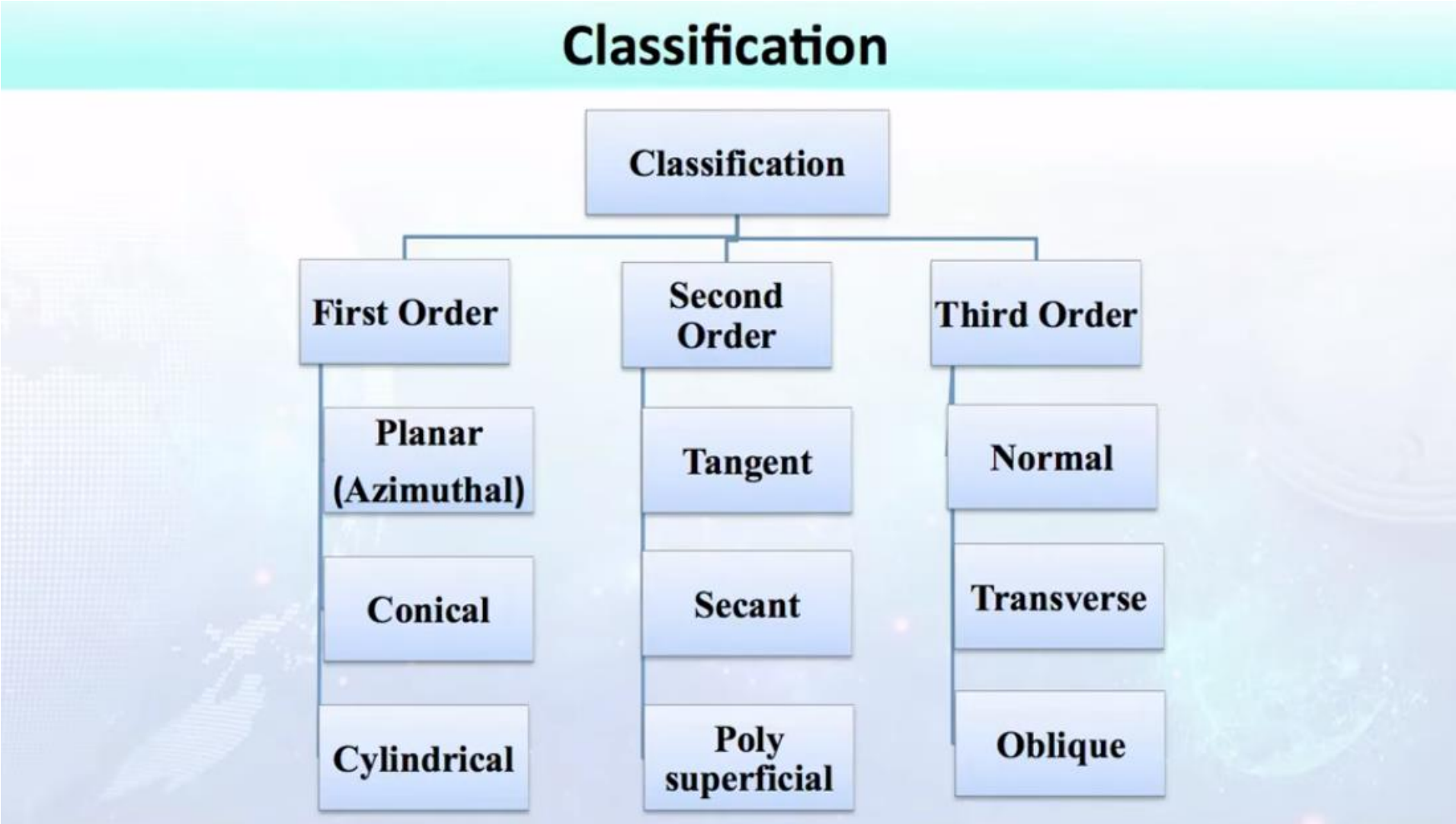
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PROJECTION SYSTEM

PROJECTION SYSTEM

- ☐ Map projections are logical and methodical transformations that permit the orderly representation of the Earth's spherical graticule on a flat map.
- ☐ Mathematically speaking, map projections are transformations of geographic co-ordinates (latitude, longitude) into the Cartesian (x, y) co-ordinate space of the map.
- ☐ A map projection is the manner in which the three dimensional spherical surface of the Earth is represented on a two-dimensional surface of a piece of paper.
- ☐ The term 'projection' comes from the idea of placing a light source within a transparent globe and projecting shadows of the meridians, and parallels onto a sheet of paper placed tangent to the globe.
- ☐ It can be defined as a "systematic arrangement of intersecting lines on a plane that represent and have one to one correspondence with the meridians and parallels on the datum surface".
- ☐ Each map projection has certain strengths and weaknesses in terms of accuracy of shape, area, distance and direction.
- ☐ It is not possible for any of the projection to retain more than one of these characteristics over a large part of the earth.
- ☐ It is important to identify that because of the curvature of the earth, all map projections distort distance and directional relationships.
- ☐ The effect of curvature on the distances and directions can be safely ignored for a small area but for a larger area distortion cannot be avoided.

The classification should follow a standard pattern so that any regular (non- conventional) can be described by a set of criteria and conversely a set of criteria will define a regular projection. The classification scheme is as follows:



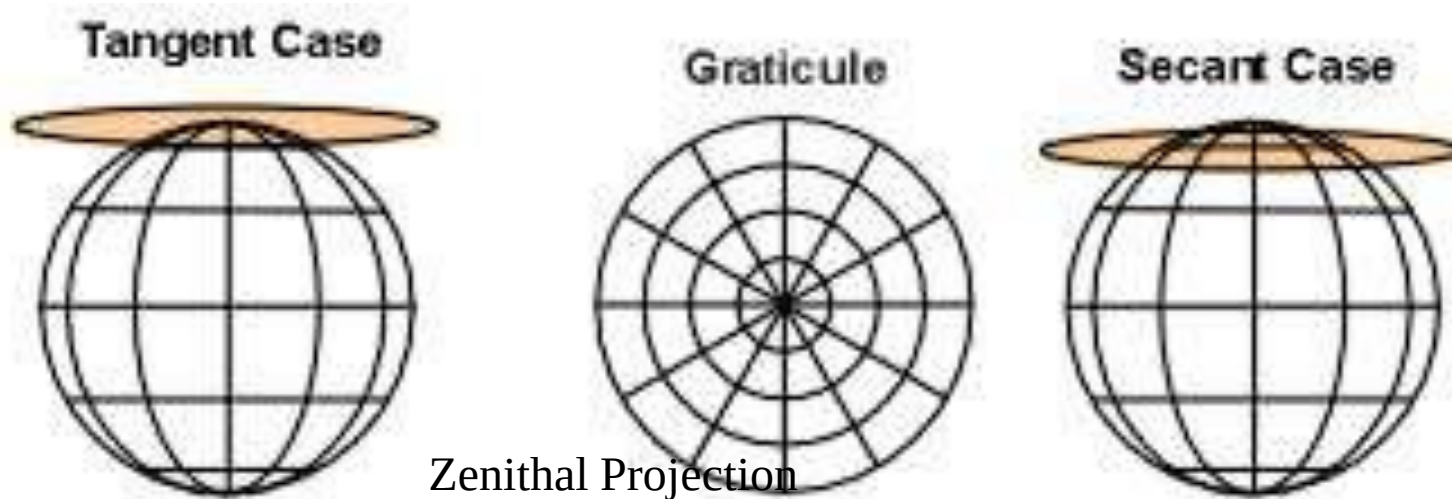
FIRST ORDER

Planar (azimuthal): Planar or azimuthal projections are accomplished by drawing lines from a given perspective point through the globe on a tangent plane.

A tangent plane intersects the global surface at only one point and is perpendicular to a line passing through the center of the sphere. Thus, these projections are symmetrical around a chosen center or central meridian.

Choice of the projection center determines the aspect, or orientation of the projection surface.

Planar projections may be centered on the poles (polar aspect), at a point on the equator (equatorial aspect), or at any other orientation (oblique aspect).

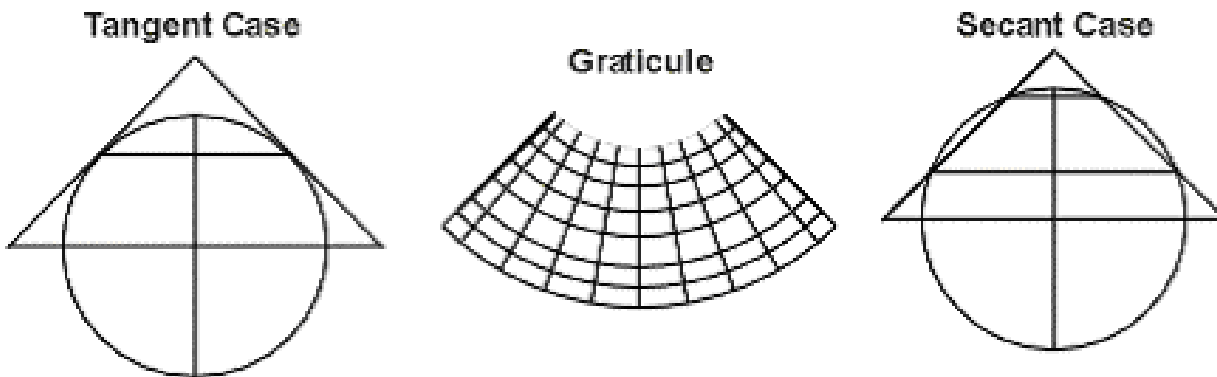


The important features of this category of projection are:

- ☐ Only a part of the earth surface is visible,
- ☐ Distortion will occur at all the four corners,
- ☐ Distance for the most part is preserved.

CONICAL:

- ❑ Conical projections are accomplished by intersecting, or touching, a cone with the global surface and mathematically projecting lines on this developable surface.
- ❑ A tangent cone intersects the earth's surface to form a circle. On this line of intersection, termed the standard parallel, the map will be relatively error-free and possess the quality of equidistance.
- ❑ Cones may also be secant, and intersect the global surface forming two circles which will possess equidistance. The word "secant", in this instance, is only conceptual, not geometrically accurate.
- ❑ As with planar projections, the conical aspect may be polar, equatorial, or oblique. Theses have been shown in the following figure.
- ❑ It has been assumed that the center, the pole and the vertex of the cone are in a straight line. The cone touches the globe at two points. These points are neither the pole the equator. It means the points are lying in between the pole and the equator.

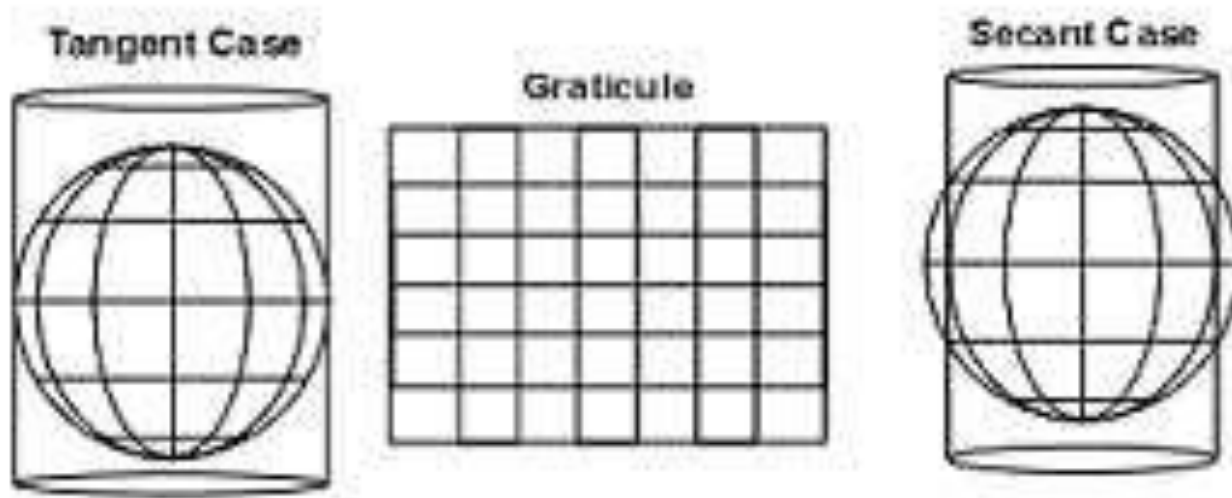


The important features are:

- ❑ Scale for the most part is preserved,
- ❑ Distance is distorted towards the bottom,
- ❑ Area is distorted.

CYLINDRICAL

- ❑ Cylindrical projections are obtained by intersecting, or touching, a cylinder with the global surface.
- ❑ The surface is mathematically projected on the cylinder, which is then “cut”, and "unrolled". If the cylinder is rotated 90 degrees from the vertical, i.e. the long axis becomes horizontal, the aspect becomes transverse and the central line of the projection becomes a chosen standard meridian.
- ❑ A tangent cylinder will intersect the global surface on only one line to form a circle as with a tangent cone. This central line of the projection is generally the equator and will possess equidistance



The important features are:

- ❑ The position of the countries near the equator is true to a large extent,
- ❑ Distortion is maximum towards the poles,
- ❑ Area for the most part is preserved.

SECOND ORDER

a) Tangent b) Secant c) Poly superficial (a series of successive projection surfaces).

These three approaches are applied to minimize the area of distortion by intersecting the globe instead of touching it. On the basis of the nature of contact to the generating globe the projections may be called as tangent, secant and polysuperficial.

A flat plane will intersect the globe along a circular line whereas a cone or cylinder along two circular lines.

Thus by intersecting the generating globe the extent of distortion free area is increased from a point to a line in case of a flat plane and from one line to two lines for cone and cylinder.

Meaning thereby there will be two standard parallels, instead of one. The distortion free area may further be increased if more than one identical shape surfaces are made tangent to the globe.

It will result into Polyconic (for successive cones) or polycylindrical (for successive cylinders) or polyhydic (successive plane) projections

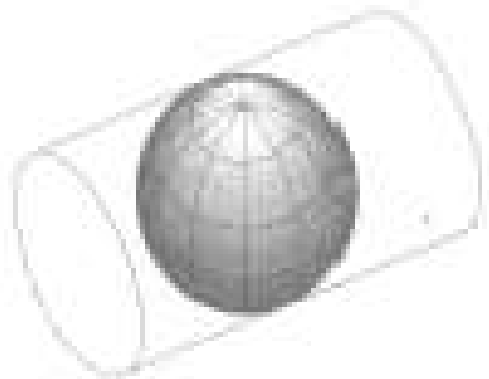
THIRD ORDER

a) Normal b) Transverse c) Oblique

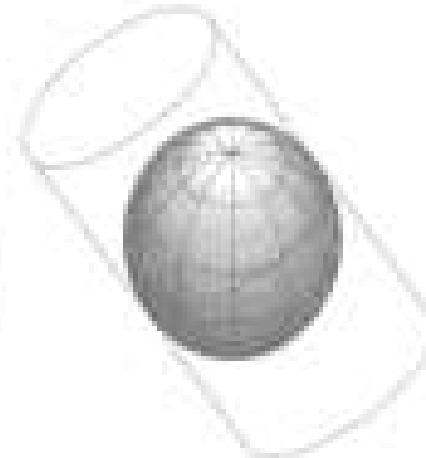
- ❑ Alignment of the generating projection surface with respect to the datum surface results to the three types of projections.
- ❑ It is said to be normal projection, if the axis of symmetry of the plane of projection is coincident with the rotational axis of the generating globe.
- ❑ If however the axis of symmetry of the projection surface is at right angles to that of the globe, a transverse case is attained.
- ❑ That is, the axis of the cylinder is at 90 to the polar axis. If the axis of symmetry of the projection surface is in between the two i.e. the two intersect at an acute angle we get an oblique case.



Normal



Transverse



Oblique

Methods of Projection

Perspective

Non Perspective

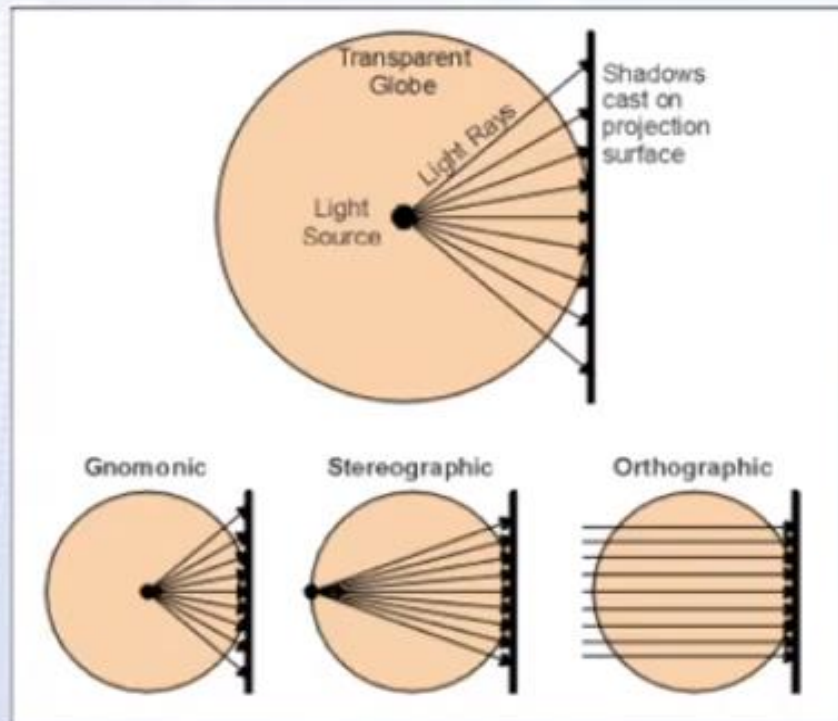


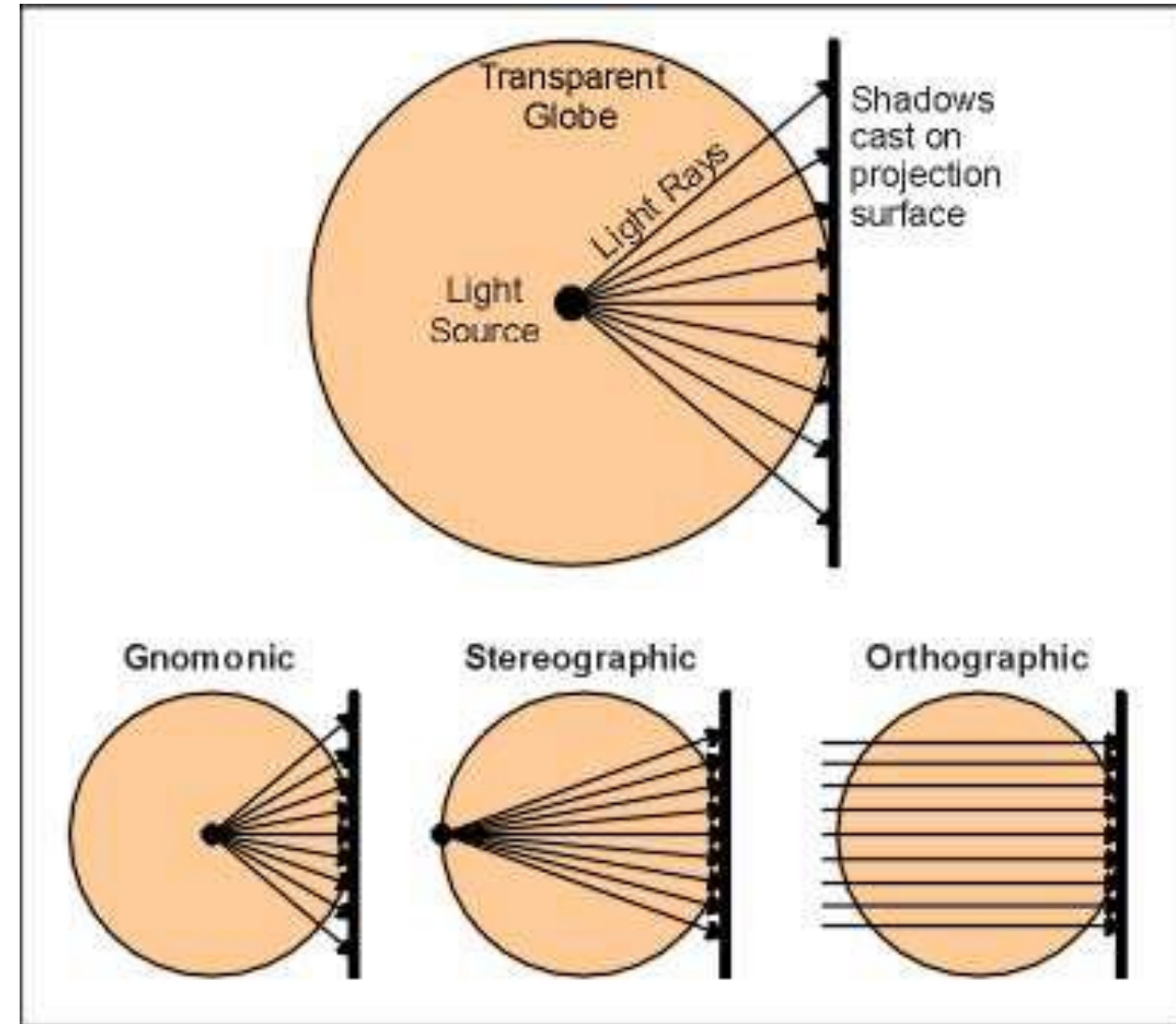
Figure 5:
Perspective
Projections

- ❑ It is a modified perspective projection for achieving certain desired properties such as true area, shape, directions and bearing.

PERSPECTIVE PROJECTIONS:

The projections in which it has been imagined that a series of rays are emerging from a source of light and they pass through a transparent globe to cast the shadow of the parallels and meridians on the projection surface. The surface may be cone, cylinder, or plane. All of these surfaces are developable meaning, they can be unfolded without distortion.

These are also known as geometric projections. Changing the position of the light source alters the pattern of parallels and meridians on the map, resulting in maps that have different geometric properties.



NON-PERSPECTIVE:

- ❑ It is a modified perspective projection for achieving certain desired properties such as true area, shape, directions and bearing.
- ❑ If the mutually intersecting lines are drawn on the projection surface without any reference to the source of light i.e. no rays or plane are involved, but are achieved by a mathematical operation. Such projections are called as non perspective.

PROPERTIES OF THE MAP PROJECTION

The classification of map projection based on the quality gives three mutually exclusive varieties representing the intrinsic geometrical properties are:

1. HOMOLOGRAPHIC (EQUAL AREA OR EQUIVALENCY)

Equivalency indicates that the areas of figures are retained on the projection surface as compared to the datum surface. This however takes place at the cost of shapes and angles getting distorted. This is used best for spatial distribution and relative sizes of spatial features, e.g. political units, population distribution, land use and natural resources etc.

2. ORTHOMORPHIC (CONFORMALITY OR TRUE SHAPE)

Conformality refers to retention of shapes or forms also retention of angles. This property is limited to close points and not the areas of significant dimensions. The angel of intersection of meridians and parallels is as it is on the globe i.e. the meridians intersect parallels at right angles, just at globe. These are also called as conformal or orthomorphic projections. These are widely used for topographic mapping and navigation purposes

3. AZIMUTHAL (TRUE BEARING)

True Direction is characterized by a direction line between two points, which crosses reference lines, e.g. meridians, at a constant angle or azimuth.

These are termed rhumb lines and this property makes it comparatively easy to chart a navigational course.

However, on a spherical surface, the shortest surface distance between two points is a great circle along which azimuths constantly change. Thus, a more desirable property may be where certain great circles are represented by straight lines.

This characteristic is most important in aviation

4. EQUIDISTANT

Equidistant means there is correct representation of distances between two points on the projection surface as compared to the distance on the datum surface.

This property is limited to a certain specified set of points and is not a general property between points on the two surfaces e.g. along central meridians in case of cylindrical projection and along standard parallels in the conical projections.

Such projections are generally used in atlas maps. Typical examples are Azimuthal Equidistant and Equidistant conic projections.

Important Map projections

UNIVERSAL TRANSVERSE MERCATOR'S PROJECTION (UTM)

- ❑ The UTM is a modified version of Mercator's projection.
- ❑ It is probably the most famous projection of all time and the best map projection for navigators because of its unique quality that an earth-bound traveler, following a line of constant compass direction, can chart the course on the Mercator projection with a Straight Line.
- ❑ This quality was not accidental; Gerardus, the developer, used a logarithmic displacement of the parallels of latitude to obtain this effect as a service to navigators.
- ❑ The UTM is a conformal projection and preserves the shape of the features such as coastlines and rivers and largely used in GIS.

Universal Transverse Mercator's Projection(UTM)

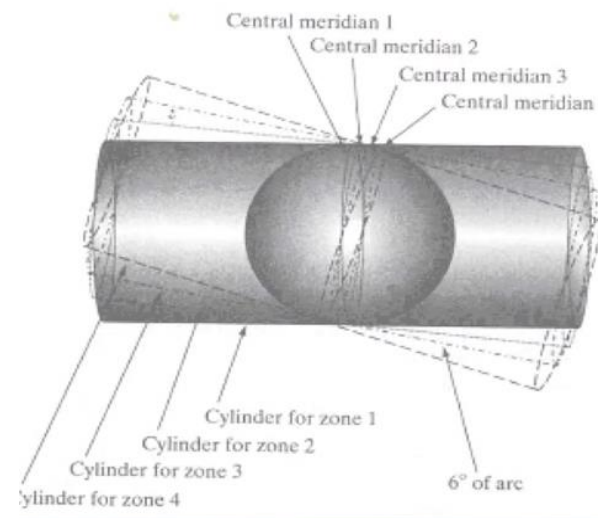


Figure 6: Transverse Mercator's Projection

- ❑ The modified transverse projection is called as Universal Transverse Mercators projection.
- ❑ It loses the properties of straight meridians and parallels of the standard Mercator projection.
- ❑ Transverse Mercator also loses the straight rhumb lines, but it is a conformal projection. Scale is true along the central meridian or along two straight lines equidistant from and parallel to the central meridian.

- ❑ It is repeatedly applied by using multiple cylinders that touch the globe at 6 intervals. The cylinder is made to intersect the globe at two standard meridians that are 180 km east and west of the central meridian.
- ❑ The UTM defines two-dimensional horizontal positions, consists of 60 zones of 6° of longitudes in width, each extending 3° east and 3° west of a central meridian extending between 80° S to 84° N latitude.
- ❑ An additional one-half degree (40-km) on each side provides for overlap into the adjacent zone. The first zone starts at 180° W to 174° W with 177° W as *central meridian*.
- ❑ The final zone, zone 60 starts at 174° E and extends east to the International Date Line.
- ❑ This projection system is largely used for large mapping projects (like the national map series produced by the U.S. Geological Survey and the Ordnance Surveys of Britain and France).
- ❑ The UTM coordinate system is based on the Transverse Mercator's projection

Universal Transverse Mercator's Projection(UTM)

Overlap in UTM

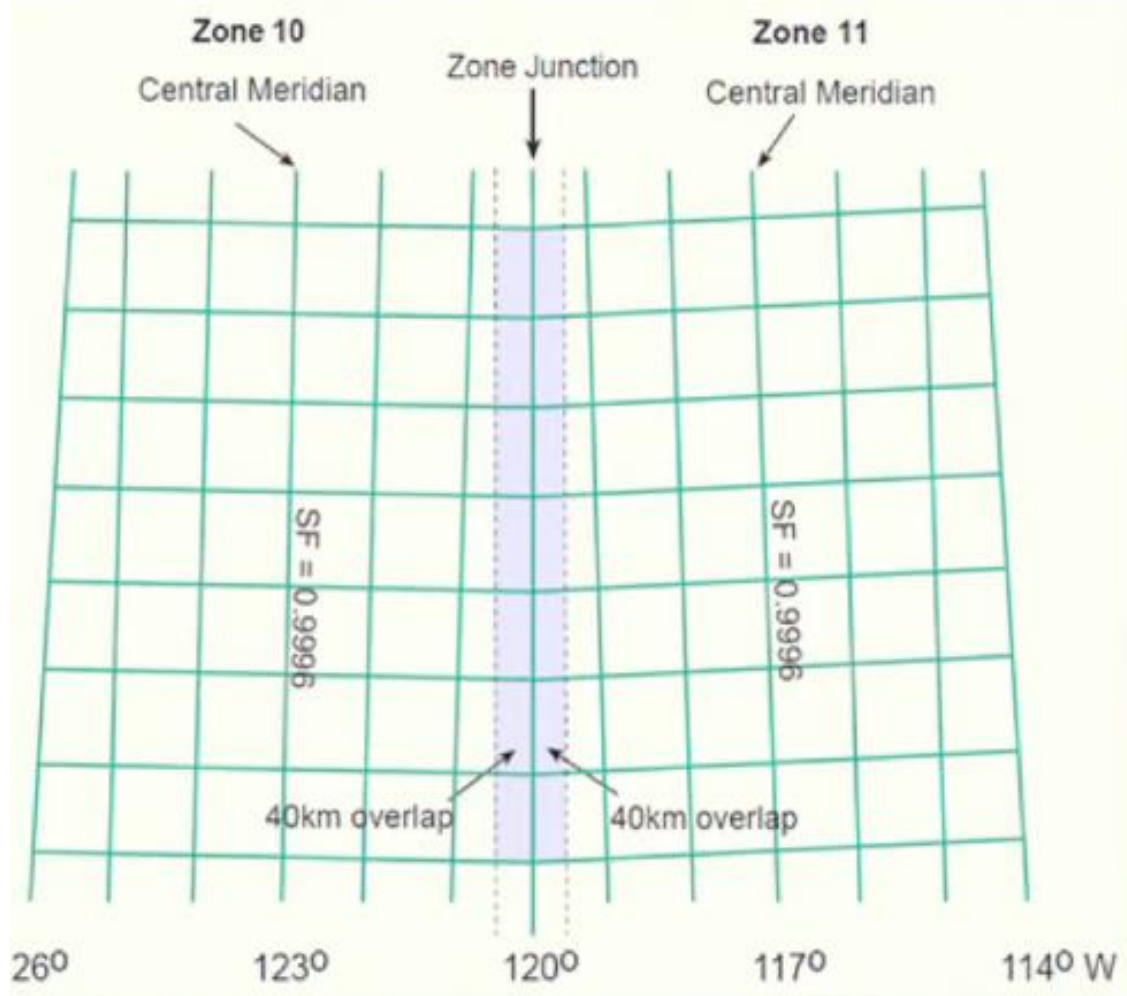


Figure 7: Overlap in UTM

Grid System

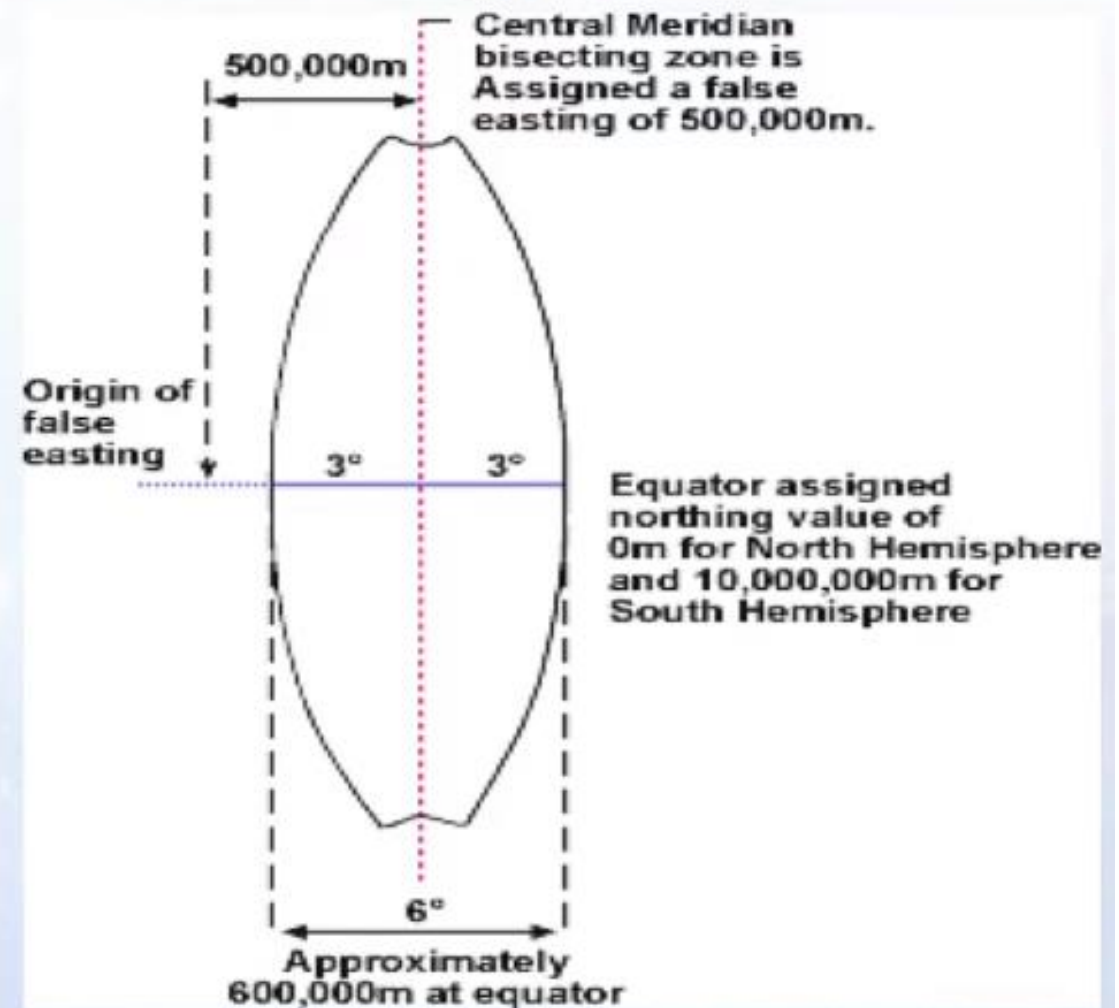


Figure 9: Universal Transverse Mercator Grid System

Universal Transverse Mercator (UTM) System

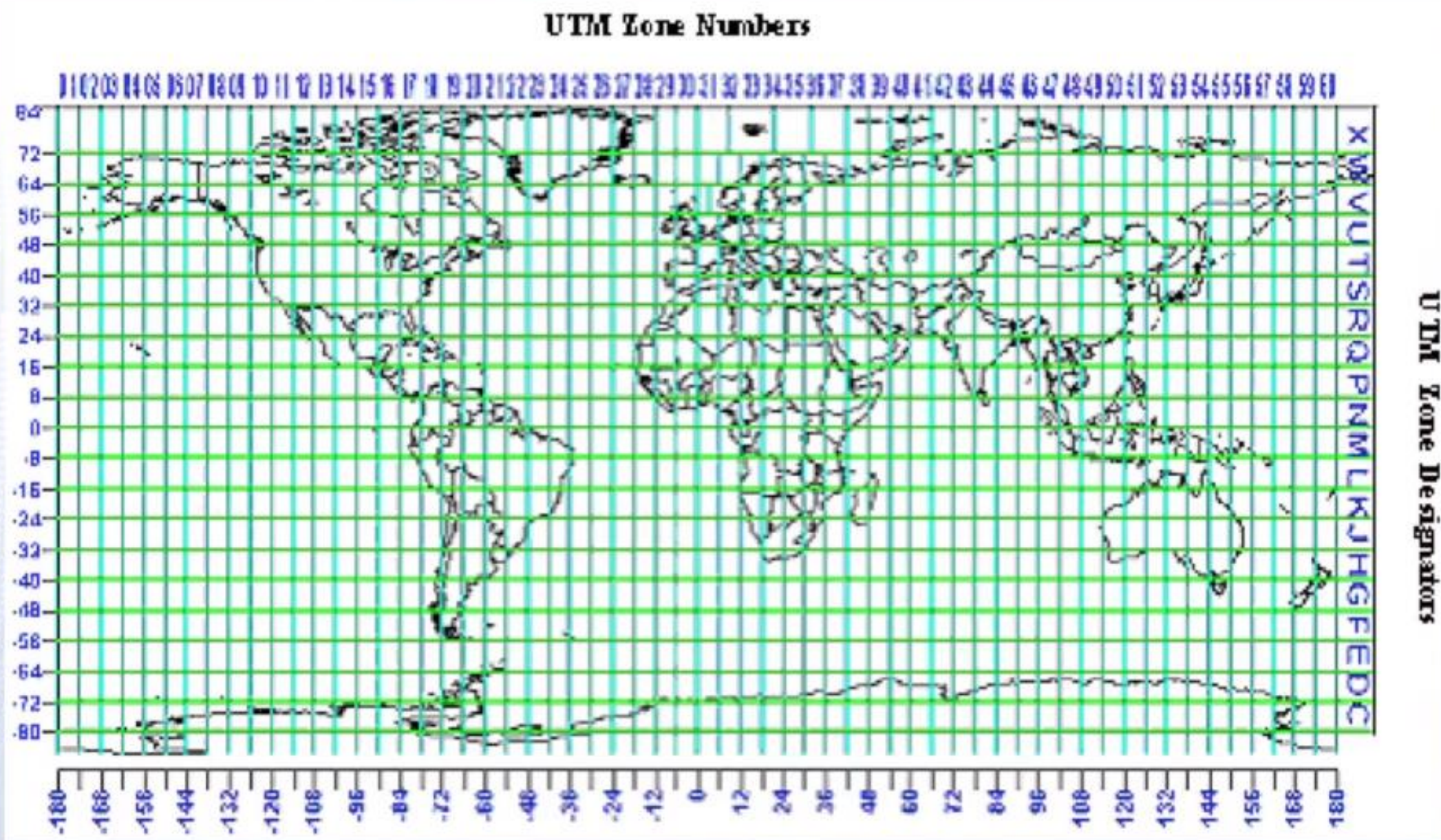


Figure 8:
Universal
Transverse
Mercator
(UTM)
System

Choice of map projection

Projection / construction	Properties	Uses
Albers equal area/ conical	Equal area; conformal along standard parallel	Small regional and national maps
Azimuth equidistant	Equidistant; true direction from center	Air and sea navigation charts, large scale maps in the equatorial and polar area
Equidistant conical	Equidistant along standard parallel and central meridian	Mid latitude areas with east-west extent, atlas mapping for small countries
Lambert Conformal Conical	Conformal; true direction locally	Navigation charts, US State Plane Coordinate System, BNG, UTM
Mercator/ Cylindrical	Conformal; True direction	Navigation charts, world maps
Polyconic/Conical	Equidistant along each parallel and central meridian	Topographic maps, USGS, SoI
Sinusoidal/cylindrical	Equal area, direction is also correct along central meridian and equator	World maps
Stereographic/planar	Conformal; true directions	Navigation charts for the polar regions
Transverse Mercator/ cylindrical	Conformal; true local directions	Topographic mapping for areas with north south extent, UTM coordinates,

Table: 2
Choice
of map
projection

Thank You